

QAMDO, China

WMO: 561370

Lat: 31.1475N

Lon: 97.1737E

Elev: 3316

StdP: 67.33

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.7	-10.2	-27.0	0.5	0.1	-24.9	0.6	0.3	6.4	6.3	5.4	6.8	0.3	270	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	27.2	12.6	25.8	12.3	24.4	11.9	14.7	22.2	14.1	21.5	13.6	20.7	2.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.5	13.7	16.7	12.0	13.2	16.1	11.5	12.8	15.8	54.6	22.3	52.5	21.6	50.7	20.9	17.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 5.3	4.3	3.5		-14.8	30.0	1.1	1.1	-15.6	30.8	-16.2	31.4	-16.8	32.0	-17.6	32.7	
(5)			DB	-16.4	15.7	0.9	1.0	-17.1	16.4	-17.6	17.0	-18.1	17.5	-18.7	18.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-0.4	1.9	5.1	8.8	12.9	16.2	17.2	16.6	14.4	9.2	4.1	0.3
	DBStd	6.79	2.65	2.59	2.50	2.41	2.90	2.70	2.27	2.15	2.37	2.97	2.33	2.35
	HDD10.0	1293	324	226	154	51	6	1	0	0	1	51	178	302
	HDD18.3	3478	582	459	411	286	168	73	50	59	119	282	428	561
	CDD10.0	888	0	0	1	14	97	187	222	206	133	28	0	0
	CDD18.3	31	0	0	0	0	1	9	13	7	1	0	0	0
	CDH23.3	546	0	0	0	1	41	157	168	134	42	3	0	0
	CDH26.7	64	0	0	0	0	2	23	22	15	3	0	0	0
(14) Wind	WSAvg	1.1	1.0	1.2	1.3	1.4	1.5	1.3	1.1	1.1	1.0	1.0	0.9	0.8
(15) Precipitation	PrecAvg	473	1	4	9	19	40	92	104	100	77	27	5	2
	PrecMax	652	8	21	24	60	103	162	197	241	157	69	22	17
	PrecMin	311	0	0	0	2	2	36	46	28	7	1	0	0
	PrecStd	91	2	5	7	13	22	34	36	53	28	19	5	4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	15.6	18.9	22.2	26.5	29.1	28.9	28.4	26.6	23.2	17.5	14.0
		MCWB	1.8	2.5	5.1	7.0	10.5	12.4	13.3	12.9	12.2	9.2	3.9	1.7
	2%	DB	11.2	13.2	16.1	19.9	24.4	26.9	26.9	26.5	24.4	20.9	15.2	11.7
		MCWB	0.2	1.4	3.5	6.2	9.4	11.9	13.3	12.9	11.8	8.6	3.0	0.5
	5%	DB	9.1	11.4	14.1	17.8	22.6	25.1	25.2	24.8	22.8	18.7	13.4	9.7
		MCWB	-0.8	0.5	2.5	5.3	8.8	11.7	13.1	12.6	11.6	7.6	2.2	-0.4
	10%	DB	7.1	9.5	12.3	16.1	20.6	23.3	23.5	23.0	20.9	16.5	11.6	7.7
		MCWB	-1.8	-0.3	1.8	4.6	8.2	11.3	13.0	12.5	11.2	6.7	1.5	-1.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	3.2	5.7	8.2	12.3	14.6	15.8	15.2	14.2	10.8	4.9	2.0
		MCDB	13.3	13.7	16.7	19.1	23.2	23.0	23.7	22.3	21.4	20.0	14.0	13.2
	2%	WB	0.4	1.9	4.1	7.0	10.7	13.7	15.0	14.5	13.3	9.7	3.7	0.7
		MCDB	10.5	11.8	14.1	17.5	20.8	21.9	22.6	21.8	20.7	18.5	12.8	11.0
	5%	WB	-0.7	0.8	3.2	6.2	9.8	13.2	14.3	13.9	12.6	8.7	2.8	-0.3
		MCDB	8.6	10.2	12.2	15.4	19.6	21.1	21.6	21.1	19.7	16.6	11.5	8.9
	10%	WB	-1.7	-0.1	2.3	5.4	9.0	12.6	13.7	13.3	12.0	7.6	1.9	-1.1
		MCDB	6.6	8.7	10.6	13.9	18.4	20.0	20.6	20.2	18.5	14.2	10.3	7.2
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	15.5	14.1	13.6	13.3	12.2	11.8	12.1	12.7	13.9	16.3	17.2
		MCDBR	19.7	19.1	17.3	17.3	16.4	15.9	15.5	16.3	16.4	17.6	19.4	19.9
		MCWBR	11.1	10.1	8.0	7.1	5.4	4.8	4.5	4.7	5.6	6.8	9.1	10.8
	5% WB	MCDBR	19.1	17.4	15.3	14.6	14.0	12.3	11.6	11.7	12.8	14.9	17.4	19.1
		MCWBR	10.8	9.3	7.2	5.9	5.1	4.1	4.0	4.0	4.7	6.1	8.3	10.5
(40) Clear-Sky Solar Irradiance	taub		0.233	0.271	0.331	0.347	0.324	0.306	0.307	0.307	0.283	0.256	0.231	0.217
	taud		2.536	2.436	2.245	2.251	2.416	2.580	2.609	2.587	2.634	2.645	2.622	2.622
	Ebn at Noon		1022	999	950	943	961	973	970	971	992	1008	1019	1031
	Edh at Noon		83	102	133	138	118	100	96	97	89	81	75	72
(44) All-Sky Solar Radiation	RadAvg		4.06	4.72	5.62	6.33	6.46	5.80	5.51	5.26	4.92	4.44	4.26	3.85
	RadStd		0.29	0.29	0.39	0.46	0.43	0.34	0.33	0.44	0.29	0.21	0.23	0.20

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.32	+0.50	-1.80	N/A	N/A	N/A	N/A	-154	+122	+20
(47) Regional (2 neighbors)		+0.30	N/A	-1.80	N/A	N/A	N/A	N/A	-156	+80	N/A

Nomenclature: See separate page